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Intel Core i7-8086K

Joining the 50K exclusive club



In honour of the 40th Anniversary of the Intel x86 architecture, Intel at COMPUTEX 2018, announced the limited edition Intel Core i7-8086K which is going to have a run of 50,000 units. It's rated to hit 5.0 GHz on Turbo with a base clock of 4.0 GHz. It's very much similar to the Intel Core i7-8700K that we've reviewed, in essence, that it's a 6-core/12-thread CPU with the exact same L2 and L3 cache sizes and even the per-core Turbo frequencies. Intel no longer releases per-core Turbo frequencies so we got these numbers from other sources. Simply put, the 8086K is a better binned Core i7-8700K.

It's the best that Coffee Lake-S has to offer until the launch of the 9th Gen Coffee Lake-S processors which are allegedly right around the corner. For being the best 6-core/12-thread that you can buy from Intel's stables, this tiny monster will set you back by \$425 or ₹35,300. So does the better binned CPU score in the benchmarks? We'll soon see but first, let's take a look at the specifications.

Specs and performance

Specification wise, the Intel Core i7-8086K is a better processor. The default multiplier being set at x40 gives it a 300 MHz gain over the Intel Core i7-8700K. The same is said for the Turbo clock. However, the per-core frequencies are what matter the most since the CPU load is more like a bell curve with most of the loads running on 2-5 cores. There are few tasks which will cause the CPU to switch to all 6-core or just one core. So are you getting the benefit of the increased clocks speeds? We'll let the benchmarks reveal that.



Price
₹35,300



PERFORMANCE.....87
VALUE.....75

In some of the more intensive multi-threaded benchmarks such as Blender, the 8086K is neck and neck with the 8700K. While in other runs, the 8700K performed better. In POV-Ray as well, the

two were producing the exact same numbers.

In Cinebench R15 and 11.5, the 8086K managed to get ahead of the 8700K by 5 points but in the multi-threaded run, it was left behind.

In WinRAR and 7-Zip's compression benchmarks, the two CPUs were neck and neck as well.

PCMark8 and PCMark10, the more comprehensive industry standards sided with the 8086K over the 8700K. In gaming, there was no difference between the two CPUs. They are exactly the same across all the games in our library such as Rise of the Tomb Raider, GTA V, and DOOM.

For most day to day work, can expect a minimal performance gain as seen in the PCMark runs.

Overclocking

The Intel Core i7-8086K has the same TDP as the Core i7-8700K and they're both unlocked CPUs. With the 8086K already hitting 5.0 GHz on Turbo, the headroom is much lower and thus you can't push the CPU more than the

8700K. The sheer fact that the 8086K is better binned puts the 8086K in a better position but being locked to a 95W TDP is the pain point. It should also be noted that with the 9th Gen launch so close, making the 8086K into a better CPU than the upcoming 9th Gen wouldn't play out well. However, a little extra headroom, say 100 W TDP, would have made the Intel Core i7-8086K a much better CPU.

Verdict

The Intel Core i7-8086K is a nice way for Intel to mark the occasion of the 40th anniversary of their x86 generation. Being a better binned chip, the Intel Core i7-8086K is like one of those golden batch CPUs that overclockers die to get their hands on. The current record for Cinebench R15 on 6 cores is held by the Core i7-8086K which went all the way to 6974 MHz. Of the top 10 rankings in the 6-core division, there are just two 8086K's right now. So folks are making do with the 8700K just fine.

What would have helped the 8086K is if the TDP had been bumped a tad higher and if all the per-core turbo frequencies were also changed even by 100 MHz. That way, the average performance gain over the 8700K would have been more noticeable than what it currently is. So if you aren't interested in the performance gain or if you prefer to not wait for the 9th Gen CPUs which are supposedly right around the corner, then you'd want to get the Intel Core i7-8086K. It's the best that Coffee Lake-S can provide right now.

—Mithun Mohandas

SPECIFICATIONS

CLOCK SPEED: 4.0 GHz | TURBO FREQUENCY: 5.0 GHz | CORES/ THREADS: 6/12 | LITHOGRAPHY: 14nm | TDP: 95W | PCIe LANES: 16 | PCIe VERSION: 3.0 | MEMORY SUPPORT: DDR4

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